

Entrepreneurial Career Trajectories: An Exploratory Life-Course Perspective

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Abstract

This exploratory study examines patterns of entrepreneurial engagement across the life course using Group-Based Trajectory Modeling and panel data from the National Longitudinal Survey of Youth 1979. We identify four distinct career trajectories—never, early-adulthood, mid-adulthood, and career-persistent entrepreneurship—and analyze differences in their timing, duration, and frequency. We then explore how early-life family capital is associated with trajectory membership and how these paths relate to psychological and financial outcomes in middle adulthood. Instead of focusing on single points of entry or exit, this study traces the full arc of entrepreneurial careers—revealing how they unfold, develop, and diverge over the course of individuals' working lives.

Keywords

career trajectory, group-based trajectory models, life-course perspective

Introduction

The relationship between age and entrepreneurship has long attracted attention in both academic research (Syed et al., 2024; Zhao et al., 2021) and popular media (Jayachandran, 2019; Munk, 2023), yet findings remain inconclusive. Some studies report an inverted U-shape, with activity peaking in mid-adulthood (Arum & Müller, 2009; Cowling, 2000; Minola et al., 2016; Williams, 2004; Zhang & Acs, 2018), reflecting accumulated experience and resources (Parker, 2018). Others find a consistently negative association, supporting the view that entrepreneurship is a young person's game and suggesting that aging reduces risk tolerance and increases preferences for stability (Bohlmann et al., 2017; Gielnik et al., 2012, 2018). Despite diverse samples and methods, these contradictory findings persist. Understanding why requires examining the analytical assumptions shared across this literature.

To understand the age-entrepreneurship relationship, prior research has examined how age relates to entrepreneurial entry (who becomes an entrepreneur and when) and exit

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dynamics (who leaves and why) as distinct transitions (Lévesque & Minniti, 2006; Wennberg et al., 2010; Zhao et al., 2021). While some studies examine how age effects vary through interactions or subgroup analyses (Minola et al., 2016; Zhang & Acs, 2018), they typically do not identify age-based trajectory patterns—distinct sequences of entry, duration, and exit across the life course. A trajectory-based approach offers two complementary advantages. First, examining sequences rather than isolated transitions allows researchers to observe career duration: whether entry at a certain age marks the beginning of a decades-long career or a brief experiment. Second, identifying age-based trajectory subgroups empirically rather than examining prespecified characteristics allows for the discovery of distinct life-course patterns of engagement that may not align with observable demographics. When individuals follow different age-related patterns of engagement over time, these trajectory-level differences may help explain why findings have varied across studies with different sample compositions.

We build on this trajectory perspective by applying life-course theory, which offers additional leverage for understanding how age relates to entrepreneurship. Life-course theory emphasizes that careers unfold as linked sequences of roles and transitions situated in a social and temporal context (Elder, 1985; Settersten, 2018). The same transition—such as starting a business—carries different meanings depending on when it occurs, what other roles an individual occupies, and how it aligns with age-graded expectations (Elder et al., 2003; Neugarten et al., 1965). Early entry typically occurs during identity formation, when individuals have fewer obligations but limited resources. Midlife entry occurs during career consolidation or reassessment, when individuals may have accumulated capital and expertise but face competing demands from family and careers. Because life stages present distinct resource conditions, developmental tasks, and opportunity structures, the same decision emerges from different circumstances and may lead to different subsequent patterns. The relationship between age and entrepreneurship may be heterogeneous rather than universal—not because individuals vary randomly, but because entrepreneurial careers are embedded in structured life-course contexts that vary systematically by timing.

If this perspective is correct, we should observe distinct patterns of entrepreneurial engagement rather than a single universal trajectory. Some individuals may enter early and sustain activity for decades, building long-term careers around self-employment. Others may delay entry until midlife but maintain engagement for shorter durations. Still others may exhibit episodic patterns, moving in and out of entrepreneurship multiple times, while many never enter. These patterns differ in the interconnected sequence of entry, duration, and exit—what Hasan and Sørensen (2011) termed entrepreneurial trajectories. Recognizing these trajectories would help reconcile contradictory findings in prior research: different studies may capture different trajectory patterns, producing divergent age effects that reflect sample composition. An inverted U-shape might emerge in samples with more midlife entrants, while negative age effects might appear in samples with more early entrants who sustain long careers.

We operationalize this perspective through trajectory analysis—examining patterned sequences of entrepreneurial engagement from early to middle adulthood. Building on Hasan and Sørensen's (2011) framework, we extend this approach by embedding it more fully within life-course theory and applying it to a broader, more diverse population. Our approach complements transition-focused research by shifting analytical focus to integrated developmental pathways (Abbott, 1997; Blair-Loy, 1999; Elder & Giele, 2009; Stovel et al., 1996). Rather than asking “who enters entrepreneurship?” or “who exits?” as separate questions, we ask “what sequences of entry, duration, and exit characterize

entrepreneurial careers?” This reframing treats timing, persistence, and exit as interconnected elements of career patterns shaped by shifting circumstances, developmental tasks, and accumulated experiences across the life course (Levinson, 1986a; Mayer, 2009; Super, 1980).

By adopting this integrated perspective, our framework extends entrepreneurial career research along three dimensions. Analytically, it shifts the unit of analysis from discrete events to developmental sequences, revealing how entry timing shapes not only who becomes an entrepreneur but also persistence and exit patterns. Theoretically, it connects entrepreneurship research more explicitly to life-course sociology and the broader career literature, which has emphasized trajectories and path dependence. By embedding decisions within life stage contexts, it explains why universal age effects have proven elusive. Empirically, it provides a framework for reconciling contradictory findings by examining whether different trajectory subgroups exist, and for linking trajectory patterns to their antecedents in early-life family capital and consequences for midlife outcomes such as well-being and economic security.

We investigate these issues using longitudinal data from the National Longitudinal Survey of Youth 1979 (NLSY79), which follows a nationally representative cohort from adolescence through middle adulthood. We apply group-based trajectory modeling (GBTM) (Nagin, 2005) to identify subgroups with similar age-dependent patterns of entrepreneurial engagement. We address three questions: (a) What distinct patterns of entrepreneurial engagement—including entry timing, duration, and persistence—emerge as individuals move from early to middle adulthood? (b) How are these trajectory patterns associated with early-life family capital? (c) How are different trajectories linked to midlife outcomes, including psychological well-being and lifetime earnings?

Given the lack of consensus in prior research and the absence of established theoretical models that account for integrated patterns across the life course, our approach is exploratory and inductive (Wennberg & Anderson, 2020). By letting the data guide identification of trajectory types and their associated contexts and outcomes, we aim to generate insights that refine theorizing about entrepreneurship as an integrated life-course process (Burton et al., 2016). In the next section, we develop a conceptual framework that distinguishes distinct entrepreneurial trajectory patterns based on entry timing, duration, and exit, providing the theoretical foundation for our analysis.

Conceptualizing Entrepreneurial Career Trajectories Across the Life Course

Transitions and trajectories are foundational concepts in the life-course framework (Elder, 1985), offering distinct but interrelated ways to understand how individuals experience change over time. Transitions refer to discrete changes in social roles or statuses—such as moving from wage employment into entrepreneurship—marking turning points that delineate “before” and “after” periods. Trajectories, by contrast, represent longer-term patterns of role occupancy characterized by their direction, duration, and sequencing. They encompass the accumulation and ordering of transitions over time, forming recognizable life-course pathways (Blair-Loy, 1999; Stovel et al., 1996). As Elder (1985) emphasizes, “transitions are always embedded in trajectories that give them distinctive form and meaning. (p. 31)” Transition into entrepreneurship at age 25, for example, carries different implications when it marks the beginning of a decades-long career versus a brief experiment before returning to wage employment.

Table 1. Conceptualizing Entrepreneurship Trajectories from Early to Middle Adulthood.

Entry	Exit		
	Early adulthood	Middle adulthood	Never
Early adulthood	Early adulthood-limited entrepreneurship (a)	Early to middle adulthood entrepreneurship (b)	Career-persistent entrepreneurship (c)
Middle adulthood	n.a.	Middle adulthood-limited entrepreneurship (d)	Middle to post-middle adulthood entrepreneurship (e)
Never	n.a.	n.a.	Never-entrepreneurship (f)

Source. Adapted from Hasan and Sørensen (2011).

Much of the literature on age and entrepreneurship has focused on discrete transitions—examining entrepreneurial entry (the likelihood of becoming self-employed) or exit (returning to wage employment or leaving the labor force) as separate, isolated events (Lévesque & Minniti, 2006; Ronstadt, 1986; Wennberg et al., 2010; Zhao et al., 2021). While these approaches provide valuable insights into how age influences specific career moments, they do not capture how transitions unfold over time as patterned careers.

A trajectory-based perspective addresses this limitation by examining career-spanning patterns of entrepreneurial activity. This perspective resonates with the growing literature on entrepreneurial careers (Burton et al., 2016; Carroll & Mosakowski, 1987; Koch et al., 2021; Merida & Rocha, 2021). Our contribution is to explicitly embed entrepreneurial career trajectories within a life-course framework, analyzing structural variation in pathways and their timing, duration, and frequency across working lives.

In the developmental life-course framework, scholars commonly distinguish among early, middle, and late adulthood (Erikson, 1963; Levinson, 1986b; Newman & Newman, 1991). Early adulthood (ages 19–40) is characterized by workforce entry, establishment of partnerships, and family formation. Middle adulthood (ages 40–65) is marked by career consolidation, parenting responsibilities, caregiving for aging parents, and awareness of physical aging. Late adulthood represents the retirement phase.

We adopt the trajectory segmentation proposed by Hasan and Sørensen (2011), which identifies patterns that capture key variations in the timing, duration, and persistence of entrepreneurial activity across early and middle adulthood. This typology (see Table 1) distinguishes trajectories based on when individuals enter entrepreneurship and how long they remain engaged, providing a framework for examining how entrepreneurial careers unfold within the life-course contexts described above.

We conceptualize six trajectory types based on entry timing and duration. Two represent persistent patterns. The Never/Never trajectory (“f” in Table 1) includes individuals who never enter entrepreneurship, remaining in traditional employment or outside the labor force. The Early/Never trajectory (“c” in Table 1) includes individuals who enter entrepreneurship early and maintain sustained engagement—what we term “career-persistent entrepreneurs.”

Four trajectories involve temporary engagement, which we term “career-limited entrepreneurs.” Among early-adulthood entrepreneurs, some exit shortly after entry (“early adulthood-limited”; “a” in Table 1) while others maintain engagement through middle adulthood before exiting (“early to middle adulthood”; “b” in Table 1). Their primary

engagement occurs during their 20s and early 30s. Early entrants may be opportunity-seeking innovators leveraging new technologies (Gielnik et al., 2012; Lévesque & Minniti, 2006) or necessity-driven entrepreneurs with limited employment options. Many face challenges, including limited financial resources, experience, and networks (Azoulay et al., 2020).

Among middle-adulthood entrepreneurs, some exit shortly after entry (“middle adulthood-limited”; “d” in Table 1) while others maintain longer engagement extending into later adulthood (“middle to post-middle adulthood”; “e” in Table 1). Their primary engagement occurs during their 40s and 50s.

While this framework illustrates *analytically* meaningful combinations of entry and exit based on prior research (Hasan & Sørensen, 2011), we adopt an exploratory approach that allows the data to reveal how individuals *actually* engage with entrepreneurship over time. This leads to our first research question:

Research Question 1 (RQ1): What distinct patterns of entrepreneurial engagement emerge across individuals’ prime working careers?

Early Life, Family Capital, and Entrepreneurial Career Trajectories

Research on entrepreneurship has long emphasized how family background shapes the initial transition into entrepreneurship (for a comprehensive review, see Parker, 2018). Far less is known about how early family conditions influence entrepreneurial *careers* as they unfold across the life course. Existing work tends to treat early family influence as a static background attribute rather than a dynamic set of resources that interact with career development and changing opportunities over time. This static view overlooks the possibility that such resources may leave enduring imprints, shaping not only the likelihood of entry into entrepreneurship but also the types of entrepreneurial careers individuals can pursue and sustain.

Drawing on Bourdieu’s (1986) framework, we conceptualize early-life family influence as the intergenerational transmission of cultural, social, and economic capital. Prior research has shown that these resources matter for entrepreneurial *entry* (Aldrich & Kim, 2007; P. H. Kim et al., 2006), but their implications for entrepreneurial *trajectories*—such as early entry and persistence, delayed entry, intermittent engagement, or early exit—remain underexplored. Because the existing literature offers limited guidance on how each form of capital shapes particular career paths, we take an exploratory approach, highlighting potential mechanisms while remaining open to diverse or unexpected patterns.

Family cultural capital is transmitted through childhood exposure to literacy-rich environments and information resources within the family home (Bourdieu, 1986; DiMaggio, 1982). Access to magazines, newspapers, and library resources during formative years shapes cognitive development, reading habits, and information-processing capabilities that influence long-term career trajectories (Sullivan, 2001; Teachman, 1987). Such resources may also influence how individuals perceive entrepreneurship as legitimate, how they imagine possible career paths, and how they persist when encountering obstacles (Rindova et al., 2009). Cultural capital could encourage early entry and long-term persistence, or support adaptive re-entry by cultivating flexibility and resilience.

Family social capital is transmitted through kinship ties and exposure to entrepreneurial role models that shape opportunities across the life course. Parents who are entrepreneurs themselves provide children not only with direct exposure to entrepreneurial practices but

also with entry points into established networks (Anderson & Miller, 2003; P. H. Kim et al., 2006). Early exposure can facilitate entry in young adulthood by fostering knowledge, confidence, and legitimacy. Alternatively, such resources may become consequential later, supporting delayed entry or re-entry in middle adulthood when individuals mobilize family connections. Family business exposure may also help sustain entrepreneurial activity through informal support and validation.

Family economic capital refers to the financial resources and security that families transmit across generations. Greater wealth allows children to access education, experiment with ventures, or recover from setbacks (Baldwin & Gorecki, 1991; Fairlie & Robb, 2007a). Economic resources might enable intermittent engagement—experimentation, exit, and re-entry without catastrophic consequences—or sustained persistence by allowing individuals to endure unprofitable periods. Conversely, limited resources may constrain options, producing early exits where economic pressures truncate entrepreneurial activity despite motivation or ability.

Taken together, these insights suggest that early-life family capital may shape not just the likelihood of entrepreneurship, but also the types of career trajectories individuals can pursue. Yet the pathways linking cultural, social, and economic resources to entrepreneurial trajectories remain underspecified. Accordingly, we pose the following research question:

Research Question 2 (RQ2): How do family-based forms of early-life cultural, social, and economic capital influence the likelihood of following different entrepreneurial career trajectories across early and middle adulthood?

Outcomes of Entrepreneurial Career Trajectories in Middle Adulthood

A life-course perspective emphasizes that career trajectories offer critical insights into later-life outcomes—insights often missed by research relying on static snapshots at single time points or across only two periods (Spilerman, 1977). We focus on midlife (around age 50) because it represents a stage when most individuals have reached career maturity and cumulative differences in earlier career paths become especially consequential. It is also when socioeconomic divergences rooted in prior trajectories tend to emerge and solidify (Adler & Stewart, 2010). At this stage, earnings typically plateau following the inverted U-shape pattern documented in labor economics (Mincer, 1974), making midlife an appropriate point to assess the cumulative economic consequences of different entrepreneurial trajectories.

We examine how variations in entrepreneurial career trajectories are associated with three outcomes in middle adulthood: (a) well-being, (b) life satisfaction, and (c) cumulative lifetime earnings. The first two capture the psychological and social consequences of entrepreneurial engagement—how individuals evaluate their lives in terms of meaning, fulfillment, and overall satisfaction. The third reflects the economic imprint of accumulated work and entrepreneurial activity. Together, these outcomes illuminate both subjective and material dimensions of entrepreneurial career paths.

Well-Being and Life Satisfaction Outcomes

Well-being and life satisfaction represent central psychological outcomes of entrepreneurial engagement that develop cumulatively across the life course through repeated role experiences and social validation. Rather than assuming uniform effects, different entrepreneurial trajectories may generate divergent outcomes depending on their timing, duration, and developmental context (Erdogan et al., 2012; Stephan, 2018). Understanding these consequences requires examining how entrepreneurial roles align with the developmental tasks and resource constraints of different life stages (Elder, 1999; Settersten, 2018).

Early-adulthood entrepreneurship coincides with identity formation and career establishment, corresponding to Erikson's (1963) stage of "intimacy versus isolation." Success during this period can enhance self-efficacy, confidence, and agency (Bandura, 1997), reinforcing entrepreneurial identity and resilience. Yet early entry also carries risks: limited resources, sparse networks, and inexperience heighten failure rates and financial stress, which may erode self-confidence and fuel unfavorable social comparisons with peers on conventional paths (Festinger, 1954). Because these experiences occur during formative years, their psychological effects may persist into midlife.

Career-persistent entrepreneurs sustain engagement across much of their working lives, producing distinctive psychological dynamics through prolonged role occupancy. Sustained practice can foster mastery, autonomy, and purpose (Sieber, 1974), while continuity may provide "narrative coherence" that integrates life stories and strengthens satisfaction (McAdams, 2001). However, persistent exposure to uncertainty, financial volatility, and decision-making burdens may generate chronic stress. Without employment-based security or structured progression, long-term entrepreneurs may face heightened risks of burnout and retirement insecurity.

Middle-adulthood entrepreneurship typically follows careers in other domains, aligning with Erikson's (1963) stage of "generativity versus stagnation." For many, entrepreneurship provides renewed purpose by enabling contribution through business creation, job provision, or innovation. The transition may foster identity reconstruction that counters stagnation (Ibarra, 1999; Pratt et al., 2006). Accumulated resources, networks, and expertise can buffer stress and increase confidence, while maturity may strengthen coping strategies. Yet not all midlife entries are opportunity-driven. For those entering after job loss or career plateaus, entrepreneurship may feel externally imposed, reducing its psychological benefits. Shorter time horizons until retirement further amplify the urgency and stress of success or failure.

These pathways illustrate that entrepreneurship enacted at different life stages yields divergent consequences depending on available resources, concurrent developmental tasks, and age-graded social expectations (Elder, 1999). While theory suggests both potential gains and strains across trajectories, systematic comparative evidence remains limited. We therefore ask:

Research Question 3 (RQ3): Which entrepreneurial trajectory is most strongly associated with well-being and life satisfaction in middle adulthood?

Lifetime Earnings (Up to Middle Adulthood)

Research has compared the financial returns of entrepreneurial and non-entrepreneurial careers (Hamilton, 2000; Koch et al., 2021; Manso, 2016), but this work typically treats entrepreneurship as a binary state rather than a dynamic process that unfolds over time.

Prior studies seldom differentiate between sustained, intermittent, or late-life entrepreneurship. As a result, we know little about how the timing, duration, and persistence of entrepreneurial engagement shape cumulative lifetime earnings by middle adulthood.

From a life-course perspective, career-persistent entrepreneurs are often assumed to benefit from the cumulative development of entrepreneurial expertise, sector-specific knowledge, and strategic acumen acquired through long-term practice (Cope, 2005; Gompers et al., 2010; Obschonka et al., 2011). These accumulated experiences may generate higher earnings through iterative learning, reputational capital, and the gradual refinement of business models and networks. However, persistence alone does not guarantee superior financial outcomes. Many entrepreneurs pursue long-term engagement for reasons other than maximizing income—such as autonomy, occupational identity, or lifestyle goals—which may lead to highly heterogeneous earnings profiles.

The earnings implications of early-adulthood entrepreneurs are similarly varied. Some individuals exit entrepreneurship early because of financial struggles or failure, while others may exit following substantial success. Successful exits may coincide with liquidity events such as acquisitions or Initial Public Offerings (IPOs), yielding significant windfalls and enabling transitions into corporate leadership or investing where entrepreneurial experience is highly valued (Campbell, 2013; Kerr et al., 2014). For these individuals, early entrepreneurial experience can create substantial lifetime earnings advantages, even if entrepreneurship itself is not sustained. However, for those who exit under duress or from under-resourced positions, the long-term effect is often reduced income and compromised future career prospects (Bruce & Schuetze, 2004; Hyytinen & Rouvinen, 2008; Mahieu et al., 2022).

The financial outcomes of middle-adulthood entrants are likewise complex. On one hand, this stage of life is often considered advantageous for launching entrepreneurial ventures. Individuals in their 40s and 50s frequently possess managerial expertise, industry-specific knowledge, and economic capital that position them well to generate income through entrepreneurship (Evans & Jovanovic, 1989; Ruef, 2010). Prior research suggests that founders entering at this stage tend to outperform their younger counterparts on average (Azoulay et al., 2020; Dahl & Sorenson, 2012; Kautonen et al., 2014). On the other hand, a later entry may limit the total accumulation of lifetime earnings from entrepreneurship, as shorter time horizons constrain the compounding of financial returns (Lévesque & Minniti, 2006). New ventures often incur early losses, making it difficult for midlife entrants to establish steady income flows before retirement (Cromie, 1991). The pressure for quick profitability may foster conservative decision-making, potentially dampening long-term growth.

These competing dynamics—accumulated experience versus time constraints, successful exits versus failures, resource advantages versus shorter horizons—underscore the need for an exploratory approach to understanding how entrepreneurial trajectories relate to lifetime earnings.

Research Question 4 (RQ4): Which entrepreneurial career trajectories are most strongly associated with higher lifetime earnings by middle adulthood?

Methods

Data and Sample

Studying the life course of entrepreneurs requires tracking individuals over lengthy periods with numerous individual and contextual measures (Aldrich & Kim, 2007). The National Longitudinal Survey of Youth 1979 (NLSY79) is ideal for this purpose, offering over 30 years of panel data.¹ It traces the career paths of a nationally representative sample of 12,686 U.S. individuals aged 14 to 22 in 1979. Respondents were interviewed annually through 1994 and biennially thereafter, with the most recent data from 2020, when their ages ranged from 55 to 64.

The NLSY79 maintained strong retention rates—approximately 80% by 2010 and 66% by 2020—providing a robust dataset for longitudinal analysis. Missing data resulted from item nonresponse within waves, skipped waves, and attrition. Item nonresponse was relatively low, with predictor variables missing less than 10% of observations, except for family income at age 18 (26% missing).

Measures

Time-Varying Entrepreneurship. Following established practice in entrepreneurship research, we use self-employment as our indicator of entrepreneurial activity (Parker, 2018). Self-employment captures both formal business ventures and informal sole proprietorships, providing an inclusive measure of entrepreneurial engagement.

The NLSY79 distinguishes between incorporated and unincorporated self-employment, allowing us to account for heterogeneity in organizational forms. Incorporated self-employment is associated with higher growth potential, greater formality, and opportunity-driven motives (Evans & Leighton, 1989; Klapper et al., 2015; Levine & Rubinstein, 2017). Unincorporated self-employment tends to reflect necessity-driven or lifestyle entrepreneurship, pursued due to limited labor market opportunities or desires for autonomy (Bregger, 1996; Levine & Rubinstein, 2017; Parker, 2018). We separately code these categories to differentiate between qualitatively distinct entrepreneurial paths.

While incorporation status might appear endogenously determined by entrepreneurial success, longitudinal evidence from this dataset suggests incorporation primarily reflects ex ante business planning rather than ex post performance. Levine and Rubinstein (2017) demonstrate that 84.5% of incorporated entrepreneurs in the NLSY79 started their ventures as incorporated entities, while 98.1% of unincorporated entrepreneurs remained unincorporated throughout their spells. Moreover, incorporated and unincorporated entrepreneurs displayed distinct characteristics as teenagers—including different cognitive skills, personality traits, and family backgrounds—before any entrepreneurial activity could influence incorporation decisions. This temporal ordering supports treating incorporation as a meaningful analytical distinction that enhances rather than compromises causal inference.

We constructed entrepreneurial career histories using dichotomous variables spanning each year from age 18 to 50. Each variable indicates whether a respondent was self-employed that year (1 = yes, 0 = no), disaggregated by incorporation status. We defined self-employment based on primary employment status in each year. This time-varying measurement captures patterns of entry, duration, and exit across the life course.

Early-Life Predictors of Trajectory Membership. To preserve temporal order, all predictors were measured at or before age 18. Consistent with established approaches (DiMaggio, 1982; Sullivan, 2001; Teachman, 1987), we define cultural capital through childhood exposure to literacy and informational materials. We created a composite index by combining three binary measures: regular availability of magazines in the home, possession of a library card by any household member, and household newspaper subscriptions. Elevated scores on this index reflect increased exposure to literacy-enriched environments linked to educational achievement and socioeconomic success.

Social capital was measured using a binary indicator of whether any parent or household family member owned a business during childhood. This represents intergenerational exposure to entrepreneurial contexts via family-based networks and role models (Aldrich & Cliff, 2003; Sørensen, 2007).

Economic capital was measured as total family income at age 18, adjusted to 2020 dollars using the Consumer Price Index and scaled in thousands. This represents material resources associated with access to educational and occupational opportunities (I. Schoon & Duckworth, 2012).

Control variables capture additional attributes that may influence trajectory group membership. At the individual level, we account for gender, race/ethnicity (Hispanic, Black, non-Hispanic non-Black), whether born in the U.S., cognitive ability (measured by Armed Forces Qualification Test in 1980), educational attainment by age 18, Rotter Internal-External Locus of Control scores, and a standardized measure of illicit behavior. Family and environmental factors include residential separation from (step)father as of 1979, urban versus nonurban residence at age 18, and geographic region (Northeast, North Central, South, West).

Career Outcomes in Middle Adulthood. We operationalize middle-adulthood outcomes as measures collected at or around age 50, representing the midpoint of middle adulthood (ages 40–65) when career trajectories have typically stabilized. These outcomes allow us to assess whether entrepreneurial trajectories are differentially associated with hedonic well-being and material security, addressing calls to examine multidimensional correlates of entrepreneurial careers (Stephan, 2018). We examine three outcomes: psychological well-being (operationalized as reverse-coded depressive symptoms), life satisfaction, and cumulative lifetime earnings.

Psychological well-being was assessed using the seven-item Center for Epidemiologic Studies Depression Scale short form, administered to respondents aged 50 and older. Items capture depressive symptoms during the prior week (appetite, concentration, mood, sleep quality, sadness, motivation), each rated on a four-point scale (0 = rarely to 3 = most of the time). We reverse-coded items so higher scores indicate greater well-being rather than greater symptomatology, then summed scores (range: 0–21). This approach has been validated in prior research on well-being in older populations (Blanchflower & Oswald, 2004; J. E. Kim & Moen, 2001).

Life satisfaction was measured using a single-item global assessment introduced in 2014, when respondents were aged 49 to 58. Respondents rated overall life satisfaction on a seven-point scale (1 = completely dissatisfied to 7 = completely satisfied). Single-item life satisfaction measures are widely used and validated in large-scale surveys (Cheung & Lucas, 2014; Diener et al., 2013). While these outcomes are measured at slightly different ages (psychological well-being at 50+, life satisfaction at 49–58), they all capture midlife functioning at or after entrepreneurial trajectories through age 50 have developed.

Lifetime earnings represent the cumulative sum of annual incomes from ages 18 to 50, adjusted to 2020 dollars using the Consumer Price Index. We log-transformed this variable to address positive skew. Following Glaeser and Kerr (2009), zero or negative values were recoded to one before transformation to retain all cases while ensuring strictly positive inputs. This preserves the full sample, including individuals with zero lifetime earnings.

Analytical Methods

Variable-centered methods examine how individual characteristics predict outcomes across an entire population, assuming homogeneous relationships. In contrast, person-centered methods identify subgroups of individuals who share similar developmental patterns, allowing relationships to vary across these subgroups (E. W. Schoon et al., 2024). Because entrepreneurial careers involve multiple transitions, feedback loops, and path dependencies, they are better understood through person-centered approaches.

Traditional variable-centered techniques have notable limitations. Regression models treat entrepreneurial entries and exits as independent events, ignoring interdependencies across time, while growth-curve models assume that all individuals follow variations around a single average trajectory. Sequence analysis offers a partial person-centered alternative but relies on subjective classification rules and provides limited tools for formal hypothesis testing (Nagin, 2005).

Finite Mixture Models (FMM) provide a more rigorous and probabilistic person-centered framework by identifying unobserved subgroups that follow distinct developmental pathways. Rather than asking “How does age affect entrepreneurship on average?” (variable-centered), FMM asks “What distinct age-based patterns of entrepreneurial engagement exist?” (person-centered). Among its main variants—Growth Mixture Modeling (GMM), Latent Class Growth Analysis, and GBTM—we adopt GBTM. GBTM maximizes within-group homogeneity, produces clearer classifications than GMM, avoids its convergence difficulties, and aligns with our theoretical premise that entrepreneurial careers represent qualitatively distinct pathways (persistent, episodic, or absent) rather than continuous variations around a single mean (see Supplemental Appendix for details).

We implemented GBTM using Mplus for initial model selection and Stata’s *traj* procedure for final estimation, modeling binary self-employment status (1 = self-employed; 0 = not) with a logit specification. We selected the optimal number of trajectory groups based on Bayesian Information Criterion (BIC), average posterior probabilities (>0.70), and substantive interpretability (Nagin, 2005). Posterior probabilities were used to assign individuals to their most likely trajectory group. Next, we assessed the association between early-life family capital and trajectory membership through multinomial logistic regression (RQ2) and evaluated how the identified trajectories relate to midlife outcomes (RQs 3–4).

Because individuals may self-select into entrepreneurial trajectories based on pre-existing characteristics that also influence later outcomes, we incorporated entropy balancing (Hainmueller & Xu, 2013) to strengthen causal inference. This preprocessing method reweights observations to achieve covariate balance between treatment and control groups. Here, treatment corresponds to membership in any entrepreneurial trajectory, and the control group consists of those following a never-entrepreneurial trajectory. Covariates were measured at or before age 18 to preserve temporal ordering (see section, “Early-Life Predictors of Trajectory Membership,” for details). This

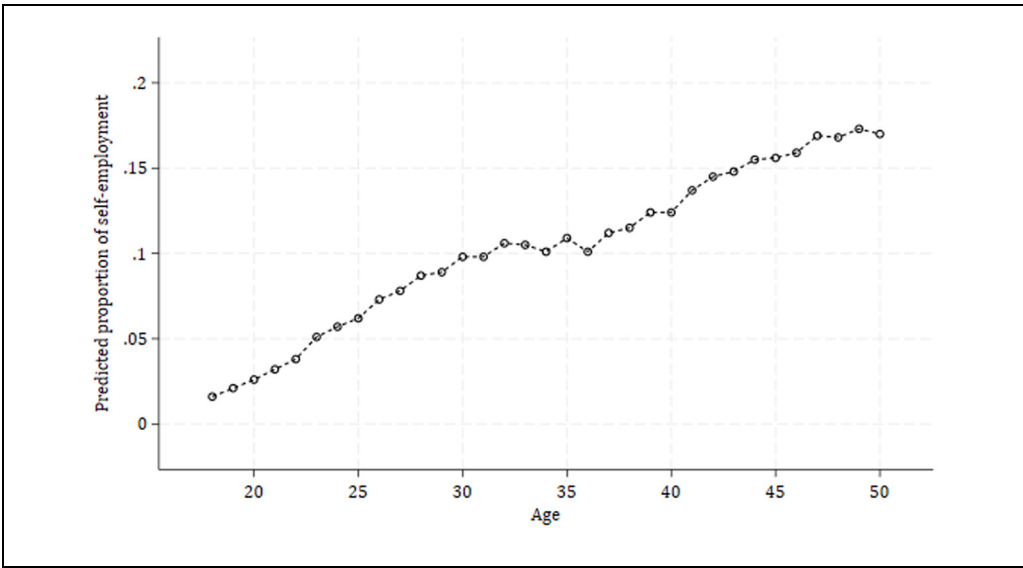


Figure 1. Growth-curve analysis of the career trajectory of entrepreneurs, age 18–50.

procedure balances a wide range of early-life antecedents linked to both trajectory selection and subsequent outcomes, thereby enhancing the credibility of estimated relationships between entrepreneurial trajectories and midlife outcomes, net of preexisting differences.

Results

What Distinct Patterns of Entrepreneurial Engagement Emerge Across Individuals' Prime Working Careers? (Research Question 1)

To answer Research Question 1, we implemented GBTM to identify distinct patterns of entrepreneurial engagement across the life course. Before presenting these trajectory-based results, we first examine what a conventional growth-curve modeling (GCM) reveals—and more importantly, what it obscures.

GCM exemplifies the variable-centered approach, assuming that all individuals follow variations around a single mean population trajectory, differing only in their starting points or rates of change (Nagin, 2005; Raudenbush, 2001). Rather than identifying distinct subgroups, GCM estimates how age affects entrepreneurship on average across the population. We specified a GCM in which the probability of self-employment varies quadratically with age. The quadratic term was statistically significant ($p < .001$), and model fit indices (Akaike Information Criterion, BIC, and sample-size adjusted BIC) favored the quadratic over the linear specification (see Table A1 in the Supplemental Appendix).

Figure 1 depicts the resulting average population trajectory. Contrary to prior studies reporting an inverted U-shape—with entrepreneurial activity peaking in mid-adulthood before declining (Lévesque & Minniti, 2006; Parker, 2018)—our results show a continuous increase throughout the observed age range. The probability of self-employment rises

Table 2. Comparative Fit Indices for GBTM Models.

Number of classes	AIC	BIC	ssBIC	Convergence	Entropy	VLMR <i>p</i> -value	aLMR <i>p</i> -value	Lowest group APP (%)	Average APP (%)
1	44,225.851	44,251.082	44,238.372	NO	N/A	N/A	N/A	100	100
2	33,406.993	33,457.457	33,432.036	YES	0.913	0.000	0.000	99	96.7
3	31,839.282	31,914.978	31,876.847	YES	0.878	1.000	1.000	90.8	93.5
4	30,301.302	30,402.229	30,351.388	YES	0.880	0.000	0.000	87.6	91.9
5	29,874.234	30,000.393	29,936.841	YES	0.883	0.216	0.216	82.2	89.3
6	29,452.525	29,603.916	29,527.654	YES	0.875	0.152	0.159	82.1	87.5
7	29,219.629	29,396.251	29,307.28	NO	0.859	0.170	0.175	80.2	86.7
8	28,960.847	29,162.701	29,061.019	NO	0.859	0.157	0.163	78.1	85.4
9	28,900.536	29,127.622	29,013.23	NO	0.824	0.004	0.004	81.3	87.4
10	28,516.473	28,768.791	28,641.689	NO	0.864	0.450	0.450	80.8	85.1

Note. APP = Average Probability of correct Placement to assigned group-based trajectory; GBTM = Group-Based Trajectory Modeling; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; ssBIC = Sample-size adjusted Bayesian Information Criterion; VLMR = Vuong-Lo-Mendell-Rubin test; aLMR = Lo-Mendell-Rubin adjusted. The bolded row indicates the model of best fit.

steadily from approximately 2% at age 18 to 17% by age 50, with gradual growth through the 20s and 30s, a brief plateau in the mid-30s, and renewed acceleration during the 40s.

Yet this aggregate pattern obscures the diversity of individual career pathways underlying it. As Nagin (2005) emphasizes, a single mean trajectory can mask profound heterogeneity in developmental processes. The steady upward trend in Figure 1 could reflect a mixture of distinct patterns—some individuals entering early and persisting, others engaging only episodically, some delaying entry until midlife, and many never entering—or compositional turnover as early entrants exit while new entrants emerge. Conventional GCM collapses these heterogeneous life-course patterns into deviations from a common mean, overlooking life-course theory's central insight that the same transition carries different meanings and consequences depending on its timing, duration, and social context (Elder et al., 2003). To uncover these qualitatively distinct pathways, a trajectory-based, person-centered approach is required.

GBTM addresses these limitations by using a finite mixture modeling framework to identify latent subgroups that share similar developmental pathways (Nagin, 2005). Rather than assuming all individuals vary around a common mean, GBTM tests whether the population comprises multiple distinct trajectory groups, each characterized by unique patterns of entry timing, duration, and persistence.

Because GBTM is an inductive approach where the number of trajectory groups is not known a priori, we incrementally tested models from 2 to 10 trajectory groups. This upper limit was based on evidence that more than 10 classes are rarely encountered or substantively meaningful in applied studies (van der Nest et al., 2020).

As shown in Table 2, models with one class and 7 to 10 classes did not converge, indicating their inability to provide reliable estimates. Our comparison, therefore, concentrated on models with two to six classes. Within this range, all classes had an Average Probability of Correct Placement above the recommended 0.70 threshold, indicating that individuals are, on average, a good fit for their assigned group (Nagin, 2005, p. 88). Similarly, all classes

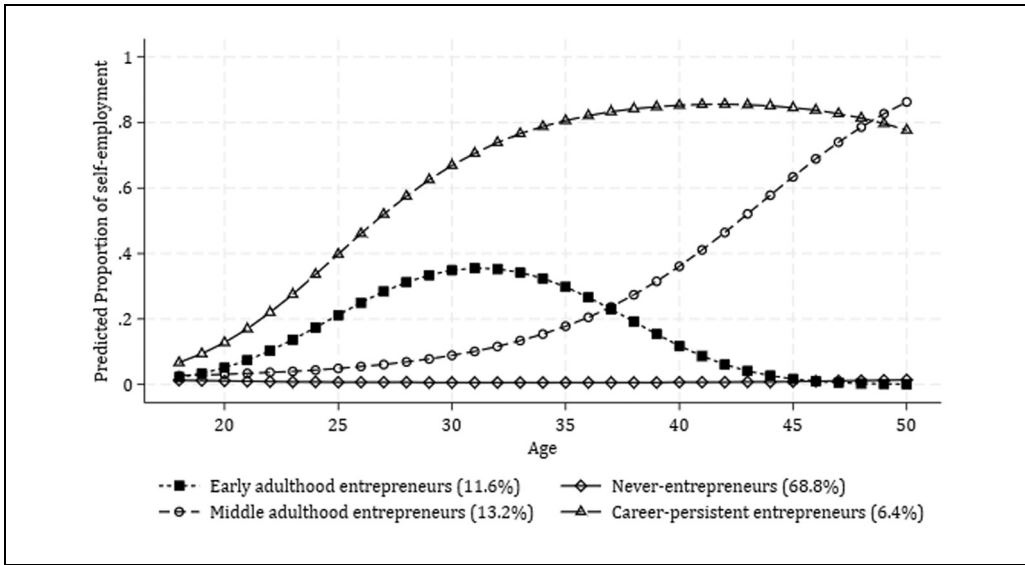


Figure 2. Four group-based career trajectories of entrepreneurs, age 18–50.

reported entropy values higher than 0.8, suggesting good classification accuracy (Muthén & Kaplan, 2004).

BIC values consistently declined with the addition of classes—a common occurrence as more parameters typically enhance data fit. Our attention, therefore, shifted to modified likelihood ratio tests (LRTs), specifically the Vuong-Lo-Mendell-Rubin test and the Lo-Mendell-Rubin adjusted LRT test (aLMR), which compare a specific model to its predecessor ($k-1$ classes). A p -value above .05 from these tests suggests that the additional class does not significantly improve model fit, negating the need for its inclusion. The 6-class model showed the best BIC value; however, the aLMR p -value for comparing the 6-class to the 5-class model was .159, indicating no significant improvement. Similarly, the aLMR p -value for comparing the 5-class to the 4-class model was .216, indicating no significant improvement with adding a fifth class. Conversely, LRTs for the 4-class model returned a p -value of .000, significantly favoring it over the 3-class model. Based on these findings, we selected the 4-class model as the best fit.

Figure 2 presents the four distinct entrepreneurial career trajectories identified by the GBTM analysis. The x-axis shows age (in years), while the y-axis indicates the proportion of individuals in each group engaged in self-employment at each age. For heuristic purposes, we have named the four groups based on their trajectory patterns.

The largest group—never-entrepreneurs (68.8%)—displays consistently low levels of self-employment throughout the observed period, remaining largely outside of entrepreneurial activity across all age points. This suggests a stable preference away from self-employment over the life course.

The career-persistent entrepreneurs (6.4%) show high levels of entrepreneurship from early adulthood through middle adulthood. This group exhibits an early and continuous rise in self-employment beginning in the late teens and early 20s, stabilizing at approximately 80% by the mid-30s and maintaining this intensity through age 50, representing a long-term commitment to entrepreneurial activity.

Two trajectories reflect career-limited entrepreneurship. Early-adulthood entrepreneurs (11.6%) exhibit a sharp increase in self-employment during the early to mid-20s, peaking in the early 30s before declining steadily, suggesting entrepreneurial activity concentrated in early adulthood that does not persist into later years. Middle-adulthood entrepreneurs (13.2%) display the opposite temporal pattern: low self-employment rates throughout their 20s, followed by a marked increase beginning in the mid-30s and continuing through age 50, reflecting delayed entry with growing engagement as individuals transition into and progress through midlife.

To further assess robustness, we conducted validation analyses using three alternative measures from the NLSY79—incorporated self-employment, unincorporated self-employment, and business ownership. Business ownership provides a more concrete organizational anchor than self-employment alone, capturing the creation, control, and management of a firm. As Light and Munk (2018) note, NLSY79 respondents distinguish between self-employment (associated with freelance or solo work) and business ownership (associated with opportunity recognition and venture development), making this distinction analytically valuable.

We also tested generalizability using the National Longitudinal Survey of Youth 1997 (NLSY97), which follows individuals born between 1980 and 1984—a younger cohort exposed to different socioeconomic, policy, and labor market conditions. Our supplemental analyses (Tables A5–A7 in the Supplemental Appendix) show that all alternative measures across both datasets, with one exception, converge on the same four entrepreneurial career trajectories. Despite differences in cohort age, macroeconomic environment, and measurement approach, both samples exhibit similarly heterogeneous patterns, indicating that the four identified trajectories represent enduring patterns in entrepreneurial engagement rather than artifacts of a particular measure, dataset, or generation.

How do Family-Based Forms of Cultural, Social, and Economic Capital Influence the Likelihood of Following Different Entrepreneurial Career Trajectories Across Early and Middle Adulthood? (Research Question 2)

To examine how family background shapes entrepreneurial career trajectories, we estimated logit and multinomial logistic regression models predicting trajectory membership from early-life family capital. Table 3 reports aggregate results comparing all entrepreneurs with never-entrepreneurs (logit) and disaggregating entrepreneurial careers into early-adulthood, middle-adulthood, and career-persistent trajectories (multinomial logit). Table 4 extends this analysis by distinguishing incorporated from unincorporated entrepreneurship, showing how specific forms of family capital channel individuals into qualitatively different organizational modes of entrepreneurial engagement.

Family cultural capital—measured through childhood access to literacy resources such as magazines, library cards, and newspapers—shows selective and enduring effects. In aggregate analyses (Table 3), cultural capital is unrelated to entrepreneurial entry overall or to early- and middle-adulthood trajectories, but demonstrates a significant positive association with career-persistent entrepreneurship ($\beta = .165, p < .05$). This pattern indicates that early exposure to cultural resources contributes less to initiating entrepreneurship and more to sustaining long-term engagement across the career span.

This pattern becomes even more pronounced when disaggregating by incorporation status (Table 4). Among incorporated entrepreneurs, cultural capital exhibits a strong and statistically significant association with career-persistent trajectories ($\beta = .568, p < .01$), but

Table 3. Logit and Multinomial Logistic Regressions of Entry Into Group-Based Trajectories.

Variables	Multinomial logistic regressions			
	Logit regression	Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)
Dependent variable	Entrepreneurs vs. Never-entrepreneurs			
Family cultural capital: magazines, library cards, & newspapers	0.053 (.181)	0.031 (.589)	0.019 (.734)	0.165* (.031)
Family social capital: Family member ever owned a business	0.429*** (.000)	0.332** (.002)	0.424*** (.000)	0.606*** (.000)
Family economic capital: Total family income, age 18 (CPI-adjusted, in thousands)	0.001 (.479)	0.001 (.638)	0.000 (.797)	0.001 (.447)
Cognitive skill	-0.003* (.078)	-0.005** (.033)	0.002 ^{ad} (.412)	-0.007** (.008)
R's education level at age 18	0.002 (.888)	0.009 (.569)	-0.011 (.496)	0.012 (.583)
External locus of control.	-0.074* (.035)	-0.034 (.509)	-0.093* (.061)	-0.110* (.095)
Illicit activity index	0.120*** (.001)	0.105* (.038)	0.147** (.002)	0.097 (.110)
Lived separately from (step)father in 1979	0.025 (.750)	0.013 (.907)	0.016 (.883)	0.062 (.679)
Lived in an urban area, age 18	0.084 (.328)	0.239* (.067)	0.003 (.982)	-0.013 (.935)
Lived in the Northeast, age 18	-0.173* (.087)	-0.204 (.176)	-0.127 (.366)	-0.215 (.264)
Lived in the North Central, age 18	-0.058 (.517)	-0.020 (.878)	-0.100 (.429)	-0.055 (.740)
Lived in the West, age 18	0.114 (.244)	0.114 (.417)	0.097 (.483)	0.143 (.429)
U.S. native (vs. Immigrants)	-0.044	-0.139	0.087	-0.088

(continued)

Table 3. (continued)

Variables	Logit regression	Multinomial logistic regressions		
Dependent variable	Entrepreneurs vs. Never-entrepreneurs	Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)
Female (vs. Male)	(.766) -0.421*** (.000)	(.495) -0.176 ^{†d} (.085)	(.690) -0.377*** ^d (.000)	(.756) -0.954*** ^{ac} (.000)
Black (vs. White)	-0.459*** (.000)	-0.560*** (.000)	-0.254 ^{†d} (.054)	-0.680*** ^c (.000)
Hispanic (vs. White)	-0.332*** (.001)	-0.312* (.036)	-0.256 [†] (.083)	-0.523*** (.009)
Constant	-0.661*** (.003)	-1.720*** (.000)	-1.619*** (.000)	-2.058*** (.000)

Note. $n = 4,633$; p -values in parentheses, two-tailed hypothesis tests. CPI = Consumer Price Index.

[†] $p < .10$. * $p < .05$. ** $p < .01$. *** $p < .001$.

^{a, c, d}Significantly different from groups labeled a through d above, $p < .05$.

Table 4. Logit and Multinomial Logistic Regressions of Entry into Incorporated versus Unincorporated Group-Based Trajectories.

Dependent variable	Incorporated entrepreneurship				Unincorporated entrepreneurship			
	Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)		Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)	
Family cultural capital	0.008 ^d (.958)	0.031 ^d (.773)	0.568*** (.009)		0.025 (.678)	0.021 (.730)	0.118 (.144)	
Family social capital: Family member ever owned a business	0.403 (.123)	0.764*** (.000)	0.398 (.156)		0.249* (.025)	0.270* (.015)	0.491*** (.000)	
Family economic capital: Total family income, age 18 (CPI-adjusted, in thousands)	0.004 (.133)	0.003* (.068)	0.003 (.226)		-0.000 (.991)	-0.001 (.606)	0.001 (.667)	
Cognitive skill	-0.002 (.710)	0.001 (.789)	0.005 (.351)		-0.005* (.027)	0.000 ^d (.885)	-0.008** (.010)	
R's education level at age 18	0.079* ^{cd} (.043)	0.018 (.548)	-0.136** (.063)		0.007 (.685)	-0.025 (.190)	0.019 (.408)	
External locus of control.	-0.123 (.361)	-0.137 (.145)	-0.068 (.635)		-0.020 (.708)	-0.074 (.174)	-0.125* (.077)	
Illicit activity index	0.059 (.640)	0.140* (.098)	0.022 (.861)		0.122* (.020)	0.143** (.006)	0.081 (.217)	
Lived separately from (step)father in 1979	-0.555 (.111)	-0.149 (.497)	-0.901* (.062)		0.139 (.247)	0.022 (.857)	0.106 (.504)	
Lived in an urban area, age 18	0.672* (.088)	-0.134 (.539)	0.624 (.137)		0.172 (.210)	-0.008 (.952)	-0.063 (.702)	
Lived in the Northeast, age 18	-0.228 (.529)	-0.258 (.321)	-0.152 (.697)		-0.154 (.342)	-0.138 (.384)	-0.255 (.223)	
Lived in the North Central, age 18	-0.130 (.691)	-0.537* (.029)	-0.316 (.384)		0.128 (.349)	-0.054 (.699)	-0.113 (.525)	
Lived in the West, age 18	-0.549 (.159)	-0.087 (.727)	-0.045 (.909)		0.174 (.246)	0.161 (.286)	0.209 (.270)	
U.S. native (vs. Immigrants)	-1.127** (.007)	-0.162 (.684)	0.161 (.833)		-0.018 (.936)	0.419 (.113)	-0.067 (.823)	

(continued)

Table 4. (continued)

Dependent variable	Incorporated entrepreneurship			Unincorporated entrepreneurship		
	Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)	Early-adulthood entrepreneurs (a) vs. Never-entrepreneurs (b)	Middle-adulthood entrepreneurs (c) vs. Never-entrepreneurs (b)	Career-persistent entrepreneurs (d) vs. Never-entrepreneurs (b)
Female (vs. Male)	-0.634 ^{a,d} (.019)	-0.567 ^{a,d} (.003)	-1.764 ^{***ac} (.000)	-0.213 ^{a,d} (.049)	-0.224 ^{a,d} (.040)	-0.832 ^{***ac} (.000)
Black (vs. White)	-0.871 ^a (.036)	-0.375 (.152)	-0.455 (.304)	-0.598 ^{***} (.000)	-0.272 ^a (.062)	-0.633 ^{***} (.001)
Hispanic (vs. White)	-0.306 (.434)	-0.293 (.292)	-0.469 (.309)	-0.282 ^a (.073)	-0.216 (.181)	-0.539 ^a (.012)
Constant	-4.245 ^{***} (.000)	-3.422 ^{***} (.000)	-4.322 ^{***} (.000)	-1.959 ^{***} (.000)	-2.027 ^{***} (.000)	-2.247 ^{***} (.000)

Note. $n = 4,633$; p -values in parentheses, two-tailed hypothesis tests. CPI = Consumer Price Index.

^a $p < .10$. ^{*} $p < .05$. ^{**} $p < .01$. ^{***} $p < .001$.

^{a, c, d} Significantly different from groups labeled a through d above, $p < .05$.

no effects for early- or middle-adulthood entrepreneurs. The advantages conferred through childhood literacy exposure—enhanced communication, analytical competence, and ease navigating formal institutions—appear particularly valuable for sustaining formal, growth-oriented ventures over extended periods. In contrast, cultural capital shows no significant relationship with unincorporated trajectories, suggesting that informal self-employment follows distinct mechanisms largely independent of early cultural advantages.

Family social capital—measured as having a family member who owned a business during adolescence—emerges as the most consistent and powerful predictor across nearly all specifications. In aggregate analyses (Table 3), adolescent exposure to family business ownership significantly increases the likelihood of entering each entrepreneurial trajectory: early-adulthood ($\beta = .332, p < .01$), middle-adulthood ($\beta = .424, p < .001$), and career-persistent ($\beta = .606, p < .001$).

Yet disaggregated analyses (Table 4) reveal striking differences in how this exposure operates across incorporated and unincorporated ventures. For incorporated entrepreneurship, the effect concentrates almost entirely among middle-adulthood entrepreneurs (.764, $p < .001$), with no significant associations for early-adulthood (.403, $p = .123$) or career-persistent (.398, $p = .156$) entrepreneurs. This pattern suggests distinct pathways into formal ventures. Early-adulthood incorporated entrepreneurs rely more heavily on formal credentials—they show significantly higher education levels by age 18 (.079, $p < .05$)—suggesting dependence on institutional networks (universities, accelerators, and venture capital) rather than family business exposure. By middle adulthood, however, individuals can combine early family exposure with accumulated professional resources—management experience, industry expertise, financial capital—to translate tacit understandings into the sophisticated competencies required for incorporated ventures. For career-persistent incorporated entrepreneurs, family cultural capital (.568, $p < .01$) appears more decisive than social capital, indicating that sustained success in formal ventures depends more heavily on literacy and analytical skills than on early business exposure.

In contrast, unincorporated entrepreneurs display a uniform and persistent pattern: adolescent family business exposure significantly predicts all three trajectories—early-adulthood ($\beta = .249, p < .05$), middle-adulthood ($\beta = .270, p < .05$), and career-persistent ($\beta = .491, p < .001$). This consistency suggests that informal ventures can directly leverage practical knowledge transmitted through family experience—such as basic business operations, self-employment norms, and customer relations—allowing individuals to activate these skills early and sustain their relevance throughout their entrepreneurial careers.

Economic capital—measured by total family income at age 18—shows limited direct effects on entrepreneurial trajectories. In aggregate analyses (Table 3), family income is not significantly associated with any entrepreneurial pathway, suggesting that heterogeneous effects across subtypes may offset one another. Disaggregated analyses (Table 4) reinforce this pattern: family income does not significantly predict incorporated entrepreneurship at any stage—early adulthood ($\beta = .004, p = .133$), middle adulthood ($\beta = .003, p = .068$), or career-persistent ($\beta = .003, p = .226$)—although coefficients remain consistently positive. For unincorporated entrepreneurship, effects are near zero and statistically nonsignificant across all trajectories, indicating that family economic resources play a minimal role in shaping either entry into or persistence within informal ventures.

Beyond family capital, individual-level characteristics exhibit distinctive associations with entrepreneurial trajectories. Cognitive skill, measured by standardized ability scores, shows trajectory-specific effects driven entirely by unincorporated ventures. Higher cognitive ability negatively predicts early-adulthood ($\beta = -.005, p < .05$) and career-persistent

($\beta = -.008, p < .05$) unincorporated entrepreneurship, while showing no significant association with incorporated trajectories. This pattern suggests a process of cognitive sorting concentrated within the informal sector, where self-employment may function as a compensatory alternative for individuals facing more limited professional or organizational opportunities.

Psychological attributes likewise differentiate across entrepreneurial trajectories. Adolescent engagement in illicit activity positively predicts early-adulthood ($\beta = .122, p < .05$) and middle-adulthood ($\beta = .143, p < .01$) unincorporated entrepreneurship, but shows no significant association with incorporated ventures. This pattern suggests that early expressions of risk tolerance or nonconformist tendencies heighten the likelihood of informal self-employment, whereas such traits are less consequential for initiating or sustaining formal business ventures.

Gender and racial disparities reveal deep and cumulative forms of stratification that intensify across entrepreneurial careers. The gender gap widens systematically over time: women experience modest disadvantages in early adulthood ($\beta = -.176, p < .10$), substantial barriers in middle adulthood ($\beta = -.377, p < .001$), and severe constraints in career-persistent entrepreneurship ($\beta = -.954, p < .001$). These disadvantages are especially pronounced within incorporated ventures, where career-persistent women face the steepest barriers ($\beta = -1.764, p < .001$).

Racial disparities follow a parallel pattern of accumulation. Black individuals face significant disadvantages across most trajectories, with particularly severe barriers in early-adulthood ($\beta = -.560, p < .001$) and career-persistent ($\beta = -.680, p < .001$) entrepreneurship. Constraints are most acute for Black entrepreneurs entering incorporated ventures in early adulthood ($\beta = -.871, p < .05$) and sustaining unincorporated ventures across the career span ($\beta = -.633, p < .001$). Hispanic individuals also encounter persistent disadvantages, most evident among career-persistent entrepreneurs ($\beta = -.523, p < .01$ overall; $\beta = -.539, p < .05$ in unincorporated ventures).

Together, these patterns underscore how family, individual, and structural forces jointly shape entrepreneurial careers. Early-life capital and personal attributes influence who enters and persists, but gendered and racialized inequalities compound over time, constraining access to formal, growth-oriented ventures and amplifying cumulative disadvantage across the entrepreneurial life course.

Which Entrepreneurial Trajectory is Most Strongly Associated With Career Outcomes in Middle Adulthood? (Research Questions 3 & 4)

To comprehensively assess how entrepreneurial trajectories shape life outcomes by midlife, we examine both psychological (well-being and life satisfaction) and economic (lifetime earnings) dimensions. All analyses employ entropy balancing to ensure comparability across trajectories on observed early-life characteristics, with successful balance documented in Table A8 (Supplemental Appendix). Results, therefore, reflect associations between entrepreneurial pathways and midlife outcomes among individuals who were otherwise similar in their developmental starting points.

Table 5 presents regression models predicting well-being and life satisfaction. The most striking finding is that aggregate effects obscure fundamental heterogeneity by incorporation status.

In aggregate analyses, self-employment appears detrimental to subjective well-being: middle-adulthood ($\beta = -.632, p < .01$) and career-persistent ($\beta = -.532, p < .05$)

Table 5. Regression Models Predicting Well-Being and Life Satisfaction in Middle Adulthood (Entropy-Balanced Samples).

Dependent variables	Well-being			Life satisfaction		
	self-employed	Incorporated self-employed	Unincorporated self-employed	Self-employed	Incorporated self-employed	Unincorporated self-employed
Never-entrepreneurs (b)						
Early-adulthood entrepreneurs (a)	reference -0.196 (.365)	reference 0.906 ⁺ (.089)	reference -0.288 ^c (.199)	reference -0.203*** (.000)	reference 0.532*** ^c (.000)	reference -0.138* (.020)
Middle-adulthood entrepreneurs (c)	-0.632** (.002)	0.077 (.829)	-0.947*** ^a (.000)	-0.110* (.047)	0.135 ^a (.163)	-0.076 (.196)
Career-persistent entrepreneurs (d)	-0.532* (.047)	0.434 (.440)	-0.680* (.016)	-0.083 (.243)	0.292* (.045)	-0.173* (.022)
Family cultural capital	0.050 (.609)	0.048 (.620)	0.047 (.626)	-0.019 (.450)	-0.020 (.448)	-0.019 (.454)
Family social capital: Family member ever owned a business	0.183 (.284)	0.173 (.312)	0.163 (.339)	-0.023 (.602)	-0.024 (.588)	-0.024 (.587)
Family economic capital: Total family income, age 18 (CPI-adjusted, in thousands)	0.001 (.618)	0.001 (.658)	0.001 (.665)	0.001* (.017)	0.001* (.029)	0.001* (.017)
Cognitive skill	0.029*** (.000)	0.029*** (.000)	0.029*** (.000)	0.000 (.987)	-0.000 (.928)	-0.000 (.960)
R's education level at age 18	0.031 (.269)	0.031 (.263)	0.029 (.296)	0.011 (.152)	0.010 (.169)	0.011 (.150)
The Rotter score	-0.305*** (.000)	-0.301*** (.000)	-0.304*** (.000)	-0.013 (.570)	-0.013 (.568)	-0.013 (.559)
Illicit activity index	-0.531*** (.000)	-0.530*** (.000)	-0.529*** (.000)	-0.102*** (.000)	-0.102*** (.000)	-0.102*** (.000)
Lived separately from (step)father in 1979	-0.816*** (.000)	-0.805*** (.000)	-0.813*** (.000)	-0.082 (.108)	-0.075 (.142)	-0.078 (.124)
Lived in an urban area, age 18	-0.360 ⁺ (.088)	-0.361 ⁺ (.087)	-0.365 ⁺ (.083)	-0.137* (.013)	-0.149** (.007)	-0.141* (.011)
Lived in the Northeast, age 18	0.421 ⁺ (.092)	0.423 ⁺ (.091)	0.415 ⁺ (.096)	-0.173** (.009)	-0.172** (.010)	-0.175** (.008)

(continued)

Table 5. (continued)

Dependent variables	Well-being		Life satisfaction	
	self-employed	Incorporated self-employed	Unincorporated self-employed	Self-employed
Lived in the North Central, age 18	0.450* (.037)	0.455* (.035)	0.455* (.034)	-0.060 (.289)
Lived in the West, age 18	0.346 (.138)	0.358 (.125)	0.360 (.123)	-0.042 (.502)
U.S. native	-0.210 (.571)	-0.188 (.613)	-0.164 (.658)	-0.034 (.726)
Female	-1.565*** (.000)	-1.533*** (.000)	-1.545*** (.000)	-0.102* (.024)
Black	0.976*** (.000)	0.980*** (.000)	0.970*** (.000)	0.158*** (.010)
Hispanic	0.358 (.160)	0.361 (.157)	0.357 (.161)	0.165* (.015)
Constant	-4.990*** (.000)	-5.271*** (.000)	-4.957*** (.000)	5.892*** (.000)
Number of Cases		4,403		4,067

Note. *p*-values in parentheses, two-tailed hypothesis tests. CPI = Consumer Price Index.

+*p* < .10. **p* < .05. ***p* < .01. ****p* < .001.

a, c. ^dSignificantly different from groups labeled a through d above, *p* < .05.

entrepreneurs report significantly lower well-being than never-entrepreneurs. Across all trajectories, entrepreneurs also exhibit lower life satisfaction, with early-adulthood entrepreneurs showing the largest deficit ($\beta = -.203, p < .001$).

Disaggregating by incorporation status, however, reveals that incorporated and unincorporated entrepreneurship produce opposite psychological outcomes. For well-being, incorporated entrepreneurs in early adulthood report higher outcomes (.906, $p < .10$) and career-persistent incorporated entrepreneurs maintain positive though nonsignificant well-being (.434, $p = .440$). By contrast, unincorporated entrepreneurs consistently report lower well-being: early-adulthood (-0.288 , ns), middle-adulthood ($-0.947, p < .001$), and career-persistent ($-0.680, p < .05$).

Life satisfaction follows the same pattern. Incorporated entrepreneurs report higher satisfaction in early adulthood (.532, $p < .001$) and career-persistent trajectories (.292, $p < .05$), while unincorporated entrepreneurs experience lower satisfaction in early adulthood ($-0.138, p < .05$) and career-persistent trajectories ($-0.173, p < .05$). Middle-adulthood effects are more muted for both types.

Overall, early entry into incorporated ventures appears especially beneficial for long-term psychological outcomes, whereas unincorporated self-employment—particularly during middle adulthood—is associated with marked psychological costs.

Table 6 presents regression models predicting lifetime earnings. Again, disaggregation by incorporation status reveals stark contrasts masked in aggregate analyses.

In aggregate, middle-adulthood entrepreneurs achieve the strongest economic returns (.161, $p < .001$), while early-adulthood entrepreneurs show no advantage ($-.013$, ns) and career-persistent entrepreneurs show positive but nonsignificant effects (.058, ns). This suggests midlife entry is particularly favorable, likely reflecting accumulated professional experience, financial resources, and networks.

Disaggregated analyses reveal that incorporated entrepreneurship consistently generates earnings premiums while unincorporated entrepreneurship provides no systematic benefit. All incorporated trajectories yield significant returns: career-persistent entrepreneurs show the highest (.679, $p < .001$), followed by middle-adulthood (.506, $p < .001$) and early-adulthood (.224, $p < .05$). These findings indicate cumulative advantages from sustained formal entrepreneurship alongside substantial midlife gains. By contrast, unincorporated entrepreneurship produces near-zero earnings associations across all trajectories: early-adulthood ($-.020$, ns), middle-adulthood (.069, ns), and career-persistent ($-.013$, ns).

Comparing across outcomes reveals that incorporated entrepreneurship enhances both earnings and psychological well-being, whereas unincorporated entrepreneurship diminishes well-being without economic compensation. The middle-adulthood incorporated trajectory appears especially advantageous, combining strong earnings returns (.506, $p < .001$) with neutral-to-positive psychological effects. The middle-adulthood unincorporated trajectory yields the most adverse profile: reduced well-being ($-.947, p < .001$) with no earnings gains.

Discussion

This study advances entrepreneurial career research by reframing entrepreneurship not as discrete transitions but as integrated career trajectories shaped by entry timing, persistence, and organizational form. This approach directly answers Burton et al.'s (2016) call to move beyond trait-based explanations toward understanding entrepreneurship as embedded in broader work histories and path-dependent sequences. Our trajectory-based framework

Table 6. Regression Models Predicting Lifetime Earnings in Middle Adulthood (Entropy-Balanced Samples).

Dependent variable	Lifetime earnings		
	Self-employed	Incorporated self-employed	Unincorporated self-employed
Never-entrepreneurs (b)	reference	reference	reference
Early-adulthood entrepreneurs (a)	-0.013 ^c (.768)	0.224 ^{*cd} (.039)	-0.020 (.665)
Middle-adulthood entrepreneurs (c)	0.161 ^{***a} (.000)	0.506 ^{***a} (.000)	0.069 (.143)
Career-persistent entrepreneurs (d)	0.058 (.294)	0.679 ^{***a} (.000)	-0.013 (.831)
Family cultural capital	0.078 ^{***} (.000)	0.074 ^{***} (.000)	0.078 ^{***} (.000)
Family social capital: Family member ever owned a business	0.116 ^{**} (.001)	0.104 ^{**} (.003)	0.118 ^{***} (.001)
Family economic capital: Total family income, age 18 (CPI-adjusted, in thousands)	0.002 ^{***} (.000)	0.002 ^{***} (.000)	0.002 ^{***} (.000)
Cognitive skill	0.013 ^{***} (.000)	0.013 ^{***} (.000)	0.013 ^{***} (.000)
R's education level at age 18	0.013 [*] (.024)	0.013 [*] (.022)	0.013 [*] (.025)
The Rotter score	-0.031 ⁺ (.083)	-0.030 ⁺ (.093)	-0.032 ⁺ (.076)
Illicit activity index	-0.028 ⁺ (.093)	-0.028 ⁺ (.097)	-0.028 ⁺ (.098)
Lived separately from (step)father in 1979	-0.233 ^{***} (.000)	-0.221 ^{***} (.000)	-0.233 ^{***} (.000)
Lived in an urban area, age 18	0.000 (.992)	-0.007 (.869)	-0.002 (.958)
Lived in the Northeast, age 18	-0.043 (.410)	-0.038 (.462)	-0.042 (.421)
Lived in the North Central, age 18	-0.052 (.248)	-0.038 (.396)	-0.053 (.242)
Lived in the West, age 18	-0.042 (.390)	-0.035 (.475)	-0.042 (.385)
U.S. native	-0.189 [*] (.013)	-0.177 [*] (.019)	-0.189 [*] (.013)
Female	-0.661 ^{***} (.000)	-0.640 ^{***} (.000)	-0.665 ^{***} (.000)
Black	-0.026 (.591)	-0.015 (.748)	-0.023 (.632)
Hispanic	0.023 (.666)	0.029 (.583)	0.023 (.666)
Constant	6.156 ^{***} (.000)	6.145 ^{***} (.000)	6.187 ^{***} (.000)

Note. $n = 4,633$; p -values in parentheses, two-tailed hypothesis tests. CPI = Consumer Price Index.

+ $p < .10$. * $p < .05$. ** $p < .01$. *** $p < .001$.

^{a, c, d}Significantly different from groups labeled a through d above, $p < .05$.

specifies the developmental mechanisms, empirical patterns, and policy implications that such a processual view entails.

Our empirical analysis yields three primary contributions. First, aggregate age effects mask fundamentally distinct subpopulations. Research debating whether entrepreneurship peaks in midlife (Lévesque & Minniti, 2006) or declines with age (Bohlmann et al., 2017) has assumed universal relationships. While our growth-curve analysis initially suggested a continuous increase through age 50, this concealed four distinct trajectories with opposite patterns—some rising, some declining, some stable. Population-averaged approaches produce conflicting results because they estimate mean effects across qualitatively different pathways, explaining why findings vary across samples with different trajectory distributions. The trajectory-based perspective reframes these inconsistencies, implying that prior contradictions may reflect differences in sample composition rather than measurement error or theoretical weakness.

Second, organizational form fundamentally differentiates entrepreneurial careers. While heterogeneity in self-employment is well established (Evans & Leighton, 1989; Levine & Rubinstein, 2017), our findings show that incorporated and unincorporated entrepreneurship represent distinct career systems rather than variations within a single domain. These systems differ not only in entry pathways but also in the types of resources and capabilities they mobilize. Incorporated careers are rooted in cultural capital and institutional competence—skills in communication, analysis, and navigating formal structures—supporting progression within growth-oriented and organizationally embedded contexts. Unincorporated careers, by contrast, rely more heavily on familial social capital and practical know-how transmitted through experience, reflecting more individualized, informally organized pathways. The divergence underscores that organizational form is not merely a legal or administrative distinction but a structural axis along which entrepreneurial careers unfold, shaping developmental trajectories, resource dependencies, and patterns of persistence over time.

Third, entry timing creates path-dependent sequences with divergent long-term consequences. Prior work on entry (Zhao et al., 2021) and exit (Wennberg et al., 2010) has illuminated discrete transitions, but examining these separately obscures how they combine into integrated patterns (Hasan & Sørensen, 2011). Early-adulthood entry produces distinct midlife psychological and economic profiles compared to middle-adulthood entry, even when both groups eventually exit. Entry timing shapes decades-long sequences and cumulative consequences in midlife, not just immediate outcomes. Early incorporated entry appears particularly beneficial because it aligns with identity formation and allows decades of compound returns, while middle-adulthood incorporated entry combines accumulated professional resources with growth-oriented structures. This extends Koch et al.'s (2021) identification of self-employment archetypes (transitory, portfolio, traditional, boundary-less) by situating patterns within life-course contexts and extends Merida and Rocha's (2021) work by showing timing effects are conditioned by organizational form.

Understanding why these patterns emerge requires examining how early-life family background, organizational form, and life-course timing interact—a mechanism–form–timing triad that structures entrepreneurial careers. Early-life family social capital—measured as parental business ownership—acts as a “permission structure,” normalizing entrepreneurship but translating into sustained engagement only when aligned with appropriate timing and organizational form. For incorporated ventures, early family exposure matters most in middle adulthood, when tacit knowledge combines with professional experience, supporting Aldrich and Cliff's (2003) family embeddedness perspective while specifying

that family influence varies by life stage. Early incorporated entrepreneurs rely more on institutional networks such as universities and accelerators (Azoulay et al., 2020), whereas career-persistent incorporated entrepreneurs depend more on cultural than social capital. For unincorporated ventures, early family exposure fosters replication with fewer complementary resources.

Family cultural capital—measured through childhood access to literacy resources—operates differently from family social capital, functioning as a “sustainability mechanism” rather than an entry facilitator. Effects concentrate specifically among career-persistent incorporated entrepreneurs, with no significant associations for early- or middle-adulthood incorporated entrepreneurs or any unincorporated trajectory. The cultural advantages from childhood literacy exposure—communication skills, analytical capabilities, and comfort navigating formal institutions—appear especially valuable for maintaining formal, growth-oriented ventures over decades (DiMaggio, 1982; Sullivan, 2001). Sustained success in incorporated ventures likely requires navigating complex regulations, managing diverse stakeholder relationships, and adapting business models to changing market conditions, all of which draw on the institutional competencies that cultural capital provides. This extends Bourdieu’s (1986) framework by specifying when and for whom cultural capital matters most in entrepreneurial careers.

Early-life family economic capital displays a distinct pattern, showing limited direct effects across most specifications. Family income neither significantly predicts incorporated entrepreneurship at any stage nor exerts a meaningful influence on unincorporated entrepreneurship.

Beyond family capital mechanisms, cumulative disadvantage processes generate pronounced stratification across entrepreneurial careers. Gender and racial disparities show that disadvantages accumulate over time and become concentrated within specific trajectory–form combinations, extending insights from entrepreneurial inequality research (Fairlie & Robb, 2007b; Jennings & Brush, 2013). The gender gap expands from marginal early-adulthood differences to substantial career-persistent constraints, particularly in incorporated ventures. This pattern is consistent with life-course principles of cumulative disadvantage (DiPrete & Eirich, 2006; Merton, 1968), in which small early disadvantages compound through feedback loops: limited access to venture capital, networks, or mentorship suppresses growth and, in turn, restricts later resource acquisition. Racial disparities exhibit similar dynamics, with Black and Hispanic entrepreneurs encountering especially severe barriers in career-persistent and incorporated trajectories—contexts where formal legitimacy and institutional access are most consequential.

These mechanisms point to broader theoretical implications for how we understand entrepreneurial careers. First, entrepreneurial careers are fundamentally heterogeneous developmental pathways rather than variations around universal processes. Life-course timing—when entry occurs relative to other developmental tasks—shapes decades-long sequences with distinct resource requirements, developmental contexts, and long-term consequences (Elder et al., 2003; Settersten, 2018). This supports moving from “variable-centered” approaches asking how age affects entrepreneurship generally toward “person-centered” approaches examining distinct developmental pathways (Nagin, 2005). Researchers should investigate whether different samples capture different trajectory distributions rather than assuming contradictory findings reflect measurement problems. The key theoretical insight is that there is no single relationship between age and entrepreneurship; rather, multiple relationships coexist within populations, each following its own developmental logic. Examining whether trajectory distributions vary systematically across

national contexts, institutional environments, or historical periods could help reconcile contradictory findings in prior literature by identifying which trajectory patterns predominate in different samples.

Second, organizational form structures fundamentally different career trajectories that may require distinct theoretical frameworks. Incorporated and unincorporated entrepreneurship create divergent career pathways with different entry mechanisms, resource dependencies, and long-term outcomes. The incorporated career path demands cultural capital, professional networks, and institutional knowledge typically accumulated through prior organizational employment. The unincorporated path depends on family social capital and local networks, requiring lower barriers to entry but limiting growth and mobility. These contrasting trajectories attract individuals at different career stages, operate through different mechanisms of career advancement, and produce different patterns of career persistence and exit. Rather than treating entrepreneurial careers homogeneously, theoretical models should specify how organizational form shapes career development from entry through midlife outcomes. Whether a unified career theory can accommodate both trajectories by identifying key contingencies remains an open question.

Third, early-life family capital effects operate through temporally specific mechanisms unfolding over decades. Different capital forms enable particular trajectory types and sustain engagement across life stages through distinct processes. Cultural capital functions as sustainability infrastructure for long-term incorporated ventures, social capital provides permission and normalization across contexts (Aldrich & Kim, 2007). The theoretical challenge is understanding not whether family background matters, but how different resources become consequential at different career moments for different organizational forms. This temporal specificity suggests that theories treating family background as uniform entry predictor miss critical heterogeneity in how capital operates across the life course. Important questions remain about whether capital forms interact—for instance, whether social capital becomes more consequential when combined with cultural or economic resources—and whether the timing of capital availability shapes its effects on entrepreneurial trajectories.

These theoretical insights have direct implications for policy and practice. By tracing careers from entry to midlife outcomes using long-term longitudinal data, this study bridges micro-level individual choice analyses with macro-level inequality and economic participation patterns, addressing calls for exploratory approaches to entrepreneurship research (Wennberg & Anderson, 2020). The trajectory perspective reveals that if entrepreneurial careers follow stratified pathways differentiated by timing, persistence, and organizational form, interventions require tailoring rather than uniform approaches.

Early-stage entrepreneurship programs should emphasize incorporation pathways and growth orientation, given that incorporated trajectories consistently generate earnings premiums and psychological benefits, while unincorporated paths often provide neither. Programming should help aspiring entrepreneurs understand the long-term implications of organizational form choices and facilitate incorporation for those with growth aspirations. Support should include navigating legal structures, accessing capital markets, and building scalable business models rather than simply encouraging self-employment (Ruef, 2010).

Midlife career transition programs should address the distinct challenges and opportunities of middle-adulthood entry. While middle-adulthood incorporated entrepreneurs achieve strong outcomes by combining accumulated resources with growth-oriented structures (Azoulay et al., 2020; Kautonen et al., 2014), middle-adulthood unincorporated entrepreneurs face the worst profile—reduced well-being without earnings compensation. Programs should screen for necessity-driven versus opportunity-driven motives, facilitate

incorporation for those with viable growth prospects, and provide alternative employment pathways for those entering unincorporated self-employment due to labor market constraints rather than entrepreneurial aspirations.

Equity initiatives should focus on removing barriers throughout entrepreneurial careers rather than applying uniform programs that presume homogeneous needs. The widening gender gap—from marginal early-adulthood disadvantages to pronounced career-persistent constraints—suggests that interventions must address compounding disadvantages at multiple stages (Jennings & Brush, 2013). Early-stage efforts should expand access to mentorship, professional networks, and venture capital, while mid-career programs should target growth bottlenecks and resource constraints that transform minor disparities into enduring inequalities. In incorporated ventures, where structural legitimacy and institutional access are central, policies should prioritize reducing barriers created by gatekeeping and financing institutions. Addressing persistent racial inequities requires similar attention to systemic obstacles within formal capital markets and legitimacy-granting processes (Fairlie & Robb, 2007b).

Limitations and Conclusion

Our findings should be interpreted within several constraints. First, we cannot fully isolate treatment effects from selection processes. While entropy balancing improves group comparability, unobserved factors that shape both trajectory entry and later outcomes likely remain. Entrepreneurial trajectories unfold through cumulative, self-reinforcing processes—not discrete treatments—limiting strictly causal interpretation. Second, our trajectory classification necessarily reduces continuous variation into discrete categories. Individuals within the same trajectory differ in timing, intensity, and specific experiences. High posterior probabilities and consistent results across alternative specifications and cohorts provide confidence in our typology, and GMM comparisons confirm that findings do not depend on zero-variance assumptions. Still, our person-centered approach identifies between-group patterns rather than within-group heterogeneity, which future research could explore through finer-grained measurement. Third, sample attrition may introduce bias if dropout correlates with both trajectory membership and outcomes. Fourth, the NLSY79 cohort remains in midlife, preventing assessment of longer-term consequences such as retirement security or late-life health.

Despite these constraints, our trajectory framework reveals entrepreneurial heterogeneity that variable-centered methods obscure. By linking trajectory membership to early family origins, cumulative life-course processes, and midlife psychological and economic outcomes, we provide a foundation for mechanism-specific theory building and targeted policy design.

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Note

1. NLS public-use data are available at no cost via the Investigator (<https://www.nlsinfo.org/>).

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